Christopher Bradley (Brad) Ferguson

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Summary

Collaboration Architect and Network Engineering Leader with 20+ years of experience designing and delivering enterprise Cisco networking, security, and collaboration solutions. Proven expertise leading large-scale migrations to Webex Calling, architecting hybrid CUCM/Webex environments, deploying Contact Center and AI-driven customer experiences, and implementing secure SIP/PSTN integrations.

Experienced in presales architecture, enterprise modernization initiatives, and automation-driven infrastructure improvements using APIs and scripting. Recognized for translating complex technical requirements into scalable, business-aligned solutions across multi-site and high-availability environments.

Education

Bachelor of Science | Software Engineering Technology | The University of Southern Mississippi
Computer Science | Faulkner State Community College | Phi Theta Kappa

Certifications

- Cisco Certified Network Professional – Collaboration (CCNP - Collaboration)
- Cisco Webex Contact Center Expert
- Cisco Webex Contact Center AI Expert
- Cisco Certified Network Professional – Enterprise (CCNP – Enterprise)
- Cisco DevNet Specialist
- Cisco DevNet Associate (DEVASC)
- Cisco DevNet Class of 2020
- Cisco Certified Design Associate (CCDA)
- Cisco Certified Network Associate Routing and Switching (CCNA)
- Certified Meraki Networking Associate (CMNA)

Experience

Collaboration Architect | C Spire Business | Aug 2023 - Present

Lead architect and technical authority for enterprise Cisco Collaboration and Webex environments, supporting internal stakeholders, wholesale customers, and professional services engagements.

- Architect and support enterprise-wide Webex solutions including Webex Calling, Webex Devices (Desk Pro, Room Kit, Room Bar, Room Kit EQX), Cisco Spaces, Webex Contact Center, and Webex Connect.
- Designed and deployed customized Webex device macro solution across 39 wireless retail locations statewide, replacing standard UI with a controlled customer-facing interface featuring branded background and one-touch specialist access.
- Engineered executive and enterprise conference room builds modeled after Cisco reference architecture standards, including device design, configuration, AV integrations, and API-driven room automation (lighting and shade controls).

- Provide presales and post-sales architecture support to Wholesale and Professional Services teams, including solution design, customer discovery, technical validation, and implementation oversight.
- Led large-scale Cisco Collaboration infrastructure projects including:
    - Cisco Unified Communications Manager (CUCM) upgrades
    - Unity Connection upgrades
    - Expressway deployments
    - Unified Contact Center Express (UCCX) implementations
    - Cisco Webex video endpoints
- Architected and executed Webex Calling migrations from CUCM, supporting hybrid coexistence and phased transitions; delivered deployments exceeding 1,000+ endpoints.
- Designed and implemented Local Gateway (CUBE) architectures for SIP/PSTN connectivity into Webex Calling, including call routing interoperability between Webex and CUCM during migration phases.
- Designed and deployed Webex Contact Center solutions ranging from traditional IVR call flows to advanced AI-driven experiences leveraging Webex Connect.
- Implemented AI-powered virtual agent for a Community College, integrating secure API calls for real-time data retrieval, backend identity verification, and automated database validation workflows.

SENIOR SYSTEMS ENGINEER | C SPIRE BUSINESS | AUG 2018 - AUG 2023

Partnered with enterprise clients to design, implement, and support Cisco-based networking, security, wireless, and collaboration infrastructures across multi-site environments.

- Provided architecture guidance and hands-on implementation for Cisco Enterprise Networking, Security, Wireless, and Collaboration solutions.
- Deployed and supported Cisco Collaboration platforms including:
    - Cisco Unified Communications Manager (CUCM)
    - Cisco Unity Connection
    - Cisco Unified Contact Center Express (UCCX)
- Designed and implemented enterprise wireless environments utilizing Cisco 9800 Series Wireless LAN Controllers.
- Engineered and supported security solutions including:
    - Cisco ASA and Firepower (FPR) firewalls with Firepower Management Center (FMC)
    - Cisco Meraki MX security appliances
    - Palo Alto Networks firewalls
    - Cisco Stealthwatch (network visibility and threat detection)
    - Cisco Identity Services Engine (ISE)
- Implemented Cisco routing and switching solutions leveraging Catalyst and Nexus platforms; supported dynamic routing protocols including EIGRP, OSPF, and BGP.
- Deployed and supported Cisco Meraki full-stack solutions (firewalls, switches, and access points) for distributed enterprise and retail environments.
- Assisted with Cisco DNA Center deployments for network automation, assurance, and policy-based management.
- Led large-scale network upgrade and refresh initiatives, coordinating cross-functional teams, vendors, and stakeholders to ensure on-time and within-scope delivery.
- Developed custom automation scripts using Python, Ansible, and APIs to streamline deployment and upgrade workflows, reducing implementation time and improving operational efficiency.
- Proactively explored network automation and infrastructure-as-code methodologies to modernize traditional networking operations and enhance customer scalability.
- Create Bill of Materials (BoMs) via Cisco Commerce Workspace (CCW) to support accurate solution pricing and quoting for customer proposals.

Adjunct Instructor | Faulkner State Community College | Aug 2008 – May 2016

Delivered online instruction for undergraduate-level networking and security courses, equipping students with foundational and advanced infrastructure knowledge.

- Taught Network Security, Windows Network Infrastructure Design, and CCNA I–IV curriculum aligned with Cisco certification tracks.
- Developed and delivered course materials covering routing and switching fundamentals, TCP/IP, subnetting, VLANs, WAN technologies, network security principles, and enterprise infrastructure design.
- Guided students through hands-on lab exercises focused on Cisco routing and switching configuration, troubleshooting methodologies, and real-world network scenarios.
- Provided mentorship and technical coaching to prepare students for Cisco CCNA certification exams.
- Evaluated student performance through labs, practical exercises, and structured assessments to ensure competency in networking fundamentals.

Senior Systems Engineer | TekLinks | June 2006 – July 2018

(Acquired by C Spire)

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